Four utilities, one payment plan

Private capstone research | 7 November 2026 | No default change

Your confirmed payment amount is $250,000 per month. This comparison leads with 12 modeled monthly payments, $3M total. The due weeks approximate a calendar schedule. The research balance sheet remains $100M with 10% unsold property and zero rent/cash rate; this is not a fully calibrated personal plan.

Result: no utility winner established

On the same paths and checks, all four utilities select a 25% candidate / 75% cash-plus-fixed-property anchor blend and pass the independent final all-risk judge under both fixed scenario laws. Raw portfolios fail the separate path-loss check. In this bounded study the hard constraints and validation grid mostly dominate the utility choice. The default stays CARA.

Definitions and what they miss

Preference	Argument / definition	Parameter and limitation
CARA surplus	u(X)=-exp(-X/T)	T=$10M. Constant dollar risk tolerance; no special aversion increase as surplus nears zero.
CRRA gross	u(G)=(G/W0)^(1-g)/(1-g)	g=10. Relative aversion on gross marked wealth; payments affect constraints, not this objective. G must be positive.
HARA payment floor	u(G)=((G-L)/S0)^(1-g)/(1-g)	S0=W0-L; g=S0/T=9.7 for the confirmed plan. Matches local dollar aversion at initial wealth. Undefined at/below floor.
Kinked shortfall	u(X)=X/W0-4*max(-X/W0,0)	Below target the marginal penalty is 5x the upside slope. Risk-neutral above target; not a payment-probability guarantee.

G is terminal marked wealth, L the terminal cash value of scheduled payments, X=G-L. For g=1, power utility becomes log. HARA(G,L,g) is CRRA(X,g), up to positive scaling. If CRRA already means power utility on NET wealth with the same g, it is not a different alternative. Putting the liability floor on X again would count payments twice.

Same-path results: $250k/month

100 training paths; 200 validation and 400 final paths per batch, four independent batches. Same return arrays, entropy-pooled nominal/stressed measures, holdings, budgets and judge for every preference. Fixed validation grid; no final repair. Native gross and surplus CEs are not comparable scores. Common CE below uses cash-first net wealth and the SAME CARA risk tolerance, $10M.

Current IID mixture law

Utility	Cash	Paid all / final all-risk	Common net ledger CE
CARA surplus	76.39%	1600/1600 / PASS	$98.253M
CRRA gross	76.39%	1600/1600 / PASS	$98.253M
HARA payment floor	76.39%	1600/1600 / PASS	$98.253M
Kinked shortfall	76.71%	1600/1600 / PASS	$98.247M

Bounded annual-volatility sensitivity

Utility	Cash	Paid all / final all-risk	Common net ledger CE
CARA surplus	72.82%	1600/1600 / PASS	$98.693M
CRRA gross	72.82%	1600/1600 / PASS	$98.693M
HARA payment floor	72.82%	1600/1600 / PASS	$98.693M
Kinked shortfall	72.82%	1600/1600 / PASS	$98.693M

All selected portfolios have a reference one-sided 95% binomial lower paid-all bound of 99.8129%, conditional on the frozen fitted model and portfolio. It is not a 99% economic guarantee. Weighted EP rates are descriptive and not binomial trials. Cross-law CE differences do not establish which law forecasts reality better.

Interpretation

For the current IID law, CARA, CRRA-gross and HARA raw weights differ by less than 0.0001 percentage points; kinked shortfall differs by up to 1.72 percentage points before blending. Under bounded volatility, all four raw portfolios are almost identical for this payment plan. The tiny common-CE differences do not justify switching the default.

The selected net ledger mean is about $98.531M under the IID law and $98.986M under bounded volatility. These are fitted-model outcomes after $3M payments, not forecasts of investment profit or comparable native utility values.

Sensitivity around your payments

These are payment-size sensitivities, NOT additional obligations: 2x, 4x and 10x your confirmed $250k/month. The preference rules, seeds, risk limits, holdings and selection grid were fixed before this extra batch. No parameter was tuned to these outcomes.

Monthly amount	Annual total	IID: selected common CE	Bounded: selected common CE
$250k	$3M	$98.247M to $98.253M	$98.693M to $98.693M
$500k (2x)	$6M	$95.247M to $95.253M	$95.693M to $95.693M
$1M (4x)	$12M	$89.247M to $89.253M	$89.693M to $89.693M
$2.5M (10x)	$30M	$71.247M to $71.253M	$71.693M to $71.693M

All four utilities select the 25% candidate blend and pass the final all-risk judge at every displayed sensitivity under each law. Every selected row pays all on 1600/1600 reference final paths, with zero observed power-utility domain violations. The cash share stays about 76.4%-76.7% under IID and 72.8% under bounded volatility. This grid does not establish a maximum affordable payment.

What remains assumed

The liability schedule, holding constraints and loss budgets can limit portfolios more strongly than preference curvature. No client risk-aversion parameter was estimated. HARA gamma is mechanically matched to local dollar aversion: (W0-L)/T, not chosen to improve results. Native CE is computed on the optimizer provisioning convention, while the common comparison uses the separate cash-first execution ledger.

The model still omits realistic sale settlement, transaction costs, taxes, rent/debt and dynamic rebalancing. Finite-sample all-risk tests and a fitted-model payment bound do not validate those omissions. No default, property holding or budget was changed.

Checks

48/48 preference solves across primary checks and monthly sensitivities used CLARABEL successfully. No solver fallback or initial formulation failure occurred in these runs. The primary study took 6.80s; the additional sensitivity study took 7.74s. Independent tests cover CE inversion, local aversion matching, HARA/CRRA-surplus identity, Jensen/scale properties, invalid domains, kink slopes and three two-asset grid oracles. Dashboard science filters are a frozen snapshot, not a recomputation of edited controls.

Check stress tests, not your case

The $95M final call and $60M+$30M schedule below were invented to test failure detection. They are NOT your payments and should not be read as realistic scenarios for you.

Check stress only	Outcome under both fixed laws
$60M week13 +$30M week52	Every preference selects only the 90% cash / 10% unsold-property anchor. Final all-risk checks pass in the fitted model. Raw HARA has 2/1600 nonpositive-surplus final outcomes under each law, so its raw native CE is UNAVAILABLE.
$95M week52	No preference has a validation candidate. The 90% cash anchor misses $5M on every path. Raw HARA native CE is UNAVAILABLE: 30/1600 IID and 23/1600 bounded final domain violations. No clipping or repair.

Training positivity is not a support guarantee

Power utilities enforce a visible $1 numerical positive-domain margin on every training path at $100M initial wealth. That is not proof of admissibility over the full unbounded return law. At/below zero, final native power CE is absent and violation counts remain visible. A family name or a smooth optimizer cannot fix a missing funding resource.

Research sources

HARA domain / affine risk tolerance: Angoshtari and Leung

Utility versus shortfall-risk constraints

HARA / shortfall theory background

Power risk aversion family discussion

Convex portfolio optimization teaching reference

Convex finance methods: Boyd lecture notes

Also checked Meucci, Risk and Asset Allocation (2005), printed pp.272-274. No textbook pages redistributed. These sources support the definitions, not our particular parameters or a preference recommendation. Concave kinked utility is not S-shaped prospect theory.